



OSCE Summaries Guidebook



OSCE
Summaries



Online OSCE Summaries

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Disclaimer:

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Abdominal Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an abdominal examination on you, which would involve me having a look at your hands, face and chest, and then having a feel of your tummy. Is that ok?*

- *For this examination I will start by positioning the bed at 45 degrees, but then I'll make it flat.*
- *Would you mind removing your shirt for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* "paraphernalia of abdominal disease"

3. Hands

Action	Sign	What it can indicate
Look at nails	Tar stains	History of smoking
	Leukonychia	Hypoalbuminemia in liver cirrhosis
	Koilonychia	Iron, B12, folate deficiency.
Ask patient to put nails together	Clubbing	Liver cirrhosis Irritable bowel disease Coeliac disease
Hands outstretched, check back of hands	Tendon Xanthomata	Hyperlipidemia
Turn hands over and look at palms	Palmar erythema	Pregnancy Hyperdynamic circulation due to increased oestrogen in liver disease
	Dupuytren's contracture	Familial Liver disease (alcoholism)
	Pigmentation of palmar creases	Addison's disease
	Finger glucose monitoring marks	Diabetes

4. Wrists

- Inspect for needle marks, bruising and muscle wasting up the arm
- Assess Radial Pulse: Look for rate (bpm) + character (thready, normal, bounding) + rhythm (regular, irregular)
- Check for renal fistulae
- *Ask for patient's blood pressure*

5. Head

- Eyes

Action	Sign	What it may indicate
Ask patient to hold down one eyelid	Conjunctival pallor	Anaemia
Ask patient to look at you	Corneal arcus	Hyperlipidemia
	Kayser-Fleischer rings	Wilson's disease
	Scleral icterus	Jaundice
Ask patient to close their eyes	Xanthelasma	Hyperlipidemia Primary biliary cirrhosis

- Face

Action	Sign	What it may indicate
Ask patient to open their mouth	Hypertrophied gums	Phenytoin Tacrolimus Nifedipine
	Poor dentition	Gives route for bacteria to enter causing IE
	Spongy and bleeding gums	Scurvy Leukaemia
	Ulcers	IBD Coeliac disease
	Pigmentation	Addison's disease
Ask them to stick Out tongue	Glossitis	B12 deficiency
	Oral candidiasis	Immunocompromised Steroid use
Close mouth	Angular stomatitis	Thiamine/B12/Folate deficiency
	Cheilosis	Infection

6. Neck (JVP and Lymph nodes)

- Check the JVP: Place the patient at 45 degrees and ask them to look to the left
 - a) Mention height above sternum (>3-4 cm abnormal)
 - b) Offer hepatojugular reflux with open mouth
- Feel the lymph nodes
 - a) Around the head
 - b) Virchow's node in the left supraclavicular triangle

7. Chest and back

- Sit patient forward: check the back for scars and spider naevi (>5 significant)
- Sit patient back: Assess for gynecomastia and spider naevi
- Assess hair on chest and the axilla
- Check for acanthosis nigricans

8. Abdomen

OBSERVE: Lie the patient flat and expose from xiphisternum to pubis

- Assess the **shape of abdomen, stomas, scars, pulsation, umbilicus** and **skin for striae**.
- Cullen's sign: bruising around the umbilicus
- Grey-Turner sign: bruising in the flank

PALPATE

- **Superficial palpation:** go level with the abdomen and palpate the 9 regions, assessing for tenderness
- **Deep palpation:** re-palpate the 9 areas and assess for masses
- Palpate the **liver, spleen** and **kidneys**
- Offer to check for an abdominal aortic aneurysm

PERCUSS

- Percuss for the liver and spleen
- Bladder: percuss suprapubic region
- Ascites (shifting dullness)

AUSCULTATE

- Listen for **bowel sounds, aortic bruits, and renal bruits**

9. Legs

- Ask if patient has tenderness --> Assess for peripheral pitting edema and DVT

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Full set of observations including blood pressure
- (**Corresponding examination**) Conduct a digital rectal examination and examine the external genitalia
- (**Bedside tests**) Urine dipstick

4bloods – Would take a full blood count + U&Es



Cardiovascular Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do a cardiovascular examination on you, which would involve me having a look at your hands, face and chest, and then having a feel and listen to your chest. Is that ok?*

- *For the purposes of this examination I'm just going to position the bed at 45 degrees.*
- *Would you mind removing your shirt for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* "paraphernalia of cardiovascular disease"
- *Listen for audible clicks of replacement valves*

3. Hands

Action	Sign	What it can indicate
Look at nails	Tar stains	History of smoking
	Koilonychia	Iron, B12, folate deficiency. Anemia can exacerbate heart failure
	Splinter haemorrhage	Subacute bacterial endocarditis
Ask patient to put nails together	Clubbing	Congenital heart disease Infective endocarditis
Hands outstretched, check back of hands	Tendon Xanthomata	Hyperlipidemia
	Peripheral Cyanosis	Cardiovascular disease, poor peripheral circulation
Turn hands over and look at palms	Osler's nodes (painful)	Subacute bacterial endocarditis
	Janeway lesions (not painful)	
Feel dorsum on hands	Temperature of hands	Poor peripheral circulation
	Check capillary refill, less than 2s	

4. Wrists

- Assess Radial Pulse: Look for rate (bpm) + character (thready, normal, bounding) + rhythm (regular, irregular)
- Assess radio-radial delay by taking both pulses together --> *Offer radio-femoral delay*
- Feel for brachial pulse (Better indicator of character)
- Check for collapsing pulse:
 - Ask whether they have any pain or stiffness in their shoulder*
 - With your hand around their wrist/lower forearm muscle bulk, rapidly raise the patient's arm upwards.*
 - You are feeling for a marked bounding sensation, supposedly like the pulse slapping your hand.*
- *Ask for patient's blood pressure and pulse pressure in both arms*

5. Head

- Eyes

Action	Sign	What it may indicate
Ask patient to hold down one eyelid	Conjunctival pallor	Anaemia
Ask patient to look at you	Corneal arcus	Hyperlipidemia
Ask patient to close their eyes	Xanthlasma	

- Face

Action	Sign	What it may indicate
Look at the patient's face	Blue lips	Peripheral cyanosis
Ask patient to open their mouth	High arched palate	Marfan's syndrome
	Poor dentition	Gives route for bacteria to enter causing IE
Ask them to stick Out tongue	Glossitis	Anaemia
	Angular Stomatitis	
Touch their tongue to palate	Blue coloring	Central cyanosis

6. Neck (JVP and Carotid Pulse)

- Check the JVP: Place the patient at 45 degrees and ask them to look to the left
- c) Mention height above sternum (>3-4 cm abnormal)
- d) Offer hepatjugular reflux with open mouth
- Check carotid pulse
- c) Auscultate first for carotid bruit
- d) Palpate for volume and character

7. Chest

OBSERVE: Ask patient to put their hands on hips to view axilla

- Assess for chest deformity: Pectus excavatum (Marfan's syndrome), Pectus carinatum

Median sternotomy scar	Indicates presence of previous coronary artery bypass or cardiac valve surgery
Lateral thoracotomy scar	Indicates previous mitral valvotomy
Subclavian scar	Pacemaker

PALPATE

- Find **apex beat** – Normally found in 5th intercostal space in left mid-clavicular line
- Identify **heaves** – hand 3 times on chest to see if there is ventricular enlargement
- Identify **thrills** – fingers on 4 auscultation points of valves

AUSCULTATE

- Feel for carotid pulse at same time to know when systole and diastole are.
Aortic Valve first - right 2nd intercostal space Pulmonary valve – Left 2nd + 3rd intercostal space
Tricuspid valve – lower left sternal edge
Mitral valve – 5th intercostal space in left -mid clavicular line
Listen for radiation to the axilla or carotids
- Use murmur accentuating movements
 - a) Patient turns onto left and breathes out – Assess mitral murmur
 - b) Patient leans up and forward – BREATHE IN --> Check tricuspid and pulmonary murmur
 - c) BREATHE OUT --> Aortic murmur

8. Back

- Listen to lung bases on back: assessing for the crackles of pulmonary oedema
- Palpate for sacral oedema

9. Legs

- Ask if patient has tenderness --> Assess for peripheral pitting edema and DVT
- Look for longitudinal scars --> indicates CABT

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Full set of observations including blood pressure
- (**Corresponding examination**) Conduct a peripheral vascular examination/respiratory examination
- (**Bedside tests**) ECG

Bloods – Would take a full blood count + U&Es and cardiac markers

Imaging – Chest X-ray

Special tests – Echocardiogram



Respiratory Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do a respiratory examination on you, which would involve me having a look at your hands, face and chest, and then having a feel and listen to your chest. Is that ok?*

- *For the purposes of this examination I'm just going to position the bed at 45 degrees.*
- *Would you mind removing your shirt for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Check for respiratory distress:* tracheal tug, lip pursing, wheezing and nasal flaring
- *Observe the surroundings:* "paraphernalia of respiratory disease"

3. Hands

Action	Sign	What it can indicate
Ask patient to put nails together	Clubbing	Atelectasis Cystic fibrosis Empyema Fibrosis
Hands outstretched, check back of hands	Tar stains	History of smoking
	Peripheral Cyanosis	Cardiovascular disease, poor peripheral circulation
	Small muscle wasting	Pancoast tumour
For asterixis, get patient to tilt hands back	Fine tremor	Use of beta-2-agonists
	Flapping tremor	CO ₂ retention in type 2 respiratory failure
Feel dorsum on hands	Temperature of hands	Poor peripheral circulation
	Check capillary refill, less than 2s	

4. Wrists

- Assess Radial Pulse: Look for rate (bpm) + character (thready, normal, bounding) + rhythm (regular, irregular)
- Check if wrists are painful by light squeeze (Indicates pulmonary hypertrophic osteoarthopathy)
- Assess respiratory rate without telling patient:
 - Tachypnoea (fever/lung disease)*
 - Bradypnoea (sedation)*
- Ask for patient's blood pressure and pulse pressure in both arms

5. Head

- Eyes

Action	Sign	What it may indicate
Ask patient to hold down one eyelid	Conjunctival pallor	Anaemia
Ask patient to look at you	Corneal arcus	Hyperlipidemia
	Xanthelasma	
Inspect the eyes	Meiosis, ptosis, anhidrosis	Horner's syndrome

- Face

Action	Sign	What it may indicate
Look at the patient's face	Blue lips	Peripheral cyanosis
Ask patient to open their mouth	Oral candidiasis	Inhaled steroid use
	Cushingoid	Steroid use
	Plethoric	Carbon dioxide retention
	Butterfly rash	SLE
Touch their tongue to palate	Blue coloring	Central cyanosis

6. Neck (JVP and trachea)

- Check the JVP: Place the patient at 45 degrees and ask them to look to the left
- e) Mention height above sternum (>3-4 cm abnormal)
- f) Offer hepatojugular reflux with open mouth
- Check the position of the trachea
 - e) Tracheal deviation
 - f) Tracheal tug
- Feel for the patient's cervical lymph nodes

7. Chest

OBSERVE: Ask patient to put their hands on hips to view axilla

- Assess for chest deformity: **Pectus excavatum (Marfan's syndrome), Pectus carinatum**
- Comment on chest wall movements: symmetry, scoliosis/kyphosis

Surgical lobectomy/pulmonectomy	Indicates removal of lobe or lung
Scar above sternal notch	Indicates previous tracheostomy

PALPATE

- Find **apex beat** – Normally found in 5th intercostal space in left mid-clavicular line
- Identify **right ventricular heave** – hand on chest to see if there is ventricular enlargement
- Feel **upper chest expansion** – palms flat on chest and should move upwards
- Feel **lower chest expansion** – hands under costal margin and thumbs should move apart

PERCUSS

- Percuss the following regions on the chest wall: **supraclavicular, infraclavicular, middle, lower, round to axilla**
- Assess **tactile vocal fremitus**: Patient says 'blue balloons' as you palpate areas of the chest
 - d) Increased resonance --> Indicates consolidation
 - e) Decreased resonance --> Indicates pleural effusion

AUSCULTATE: Tell patient to breath deep through mouth and auscultate the same areas as you percussed

- Assess **vocal resonance**: Patient says 'blue balloons' when you auscultate over an area
 - f) Increased resonance --> Indicates consolidation
 - g) Decreased resonance --> Indicates pleural effusion

8. Back

- Ask patient to sit over the bedside with their arms crossed
- Assess the back including **inspection, chest expansion, percussion, and auscultation**.
- Feel for sacral oedema on the back

9. Legs

- Ask if patient has tenderness --> Assess for peripheral pitting edema and DVT

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Full set of observations including blood pressure
- (**Corresponding examination**) Conduct a cardiovascular examination
- (**Bedside tests**) Peak flow recordings, sputum sample

Bloods – Would take a full blood count + ABG

Imaging – Chest X-ray



GALS Examination

1. Introduction

- Wash Your hands
- Introduce yourself by name and role
- Check their identity – name and DOB
- Explain the procedure - why you need to do it and what does it involve
- Ask for consent
- Expose the patient appropriately
- Check if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. Can I check your name and date of birth?
I have been asked to do an examination of your muscles and joints today, which would involve me having a look at your walk and asking you to do a number of different movements with your head, arms and legs, to screen any problems you might have. Is that ok?
- Would you mind removing all of your clothes except for your underwear, please?
- And can I just check whether you are in any pain?

2. Bedside Inspection

- Observe the patient: Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- Observe the surroundings: mobility aids, Zimmer frame

3. Screening questions

- There are 3 screening questions that you might ask at the beginning of the exam.
 - a) Do you have any pain or stiffness in your joints, muscles or back?
 - b) Can you dress yourself without difficulty?
 - c) Do you have any difficulty when going up or down the stairs?

4. Gait

- Ask the patient to walk over to the wall, turn around and walk back.
- Check for symmetry, smoothness, and step height
- Observe gait cycle (heel strike, toe-off)
- Is the turn smooth and quick?

Gait pathology	Sign	What it can indicate
Antalgic gait	Stance phase abnormally shortened to reduce time on that foot	Implies pain in specific leg
Waddling gait	Patient moves their upper body forwards and drags lower leg forward	Weakness of proximal muscles of pelvis which causes weakness of gluteus muscles
Spastic gait	Stiffness in the legs and tendency to circumduct the feet	Upper Motor Neuron (UMN) lesion
Fixed flexion/hyperextended knee	The knee is in fixed flexion/knee is hyperextended	Occurs in polio as patient have quadriceps wasting
High-stepping gait	Characterized by foot-drop due to loss of dorsiflexion	Damage to the deep fibular nerve
Trendelenburg's gait (look for Trendelenburg sign)	Support the patient's outstretched arms, get them to stand on one leg. If unsupported side of pelvis drops down, then positive sign.	Indicated damage to the gluteus medius on the supported side, due to lesion of the superior gluteal nerves

5. Patient standing

OBSERVE: Ask patient to stand in the anatomical position and observe from the **front, side and back**

Action	Sign	What to look for
Front	Posture	Obvious asymmetry
	Shoulder	Shoulder bulk and symmetry
	Elbow extension	Carrying angle (normal is 5-15 degrees)
	Leg length	Leg length inequality
	Quadriceps	Wasting in chronic joint disease

	Knees	Erythema/Hyperextension
	Ankle	Swelling and erythema (seen inflammatory arthritis or sepsis)
	Feet	Hallux valgus Midfoot deformity (flat feet)
Side	Cervical spine	Assess hyperlordosis (indicates spondylolisthesis/discitis/osteoporosis)
	Thoracic kyphosis	Hyperkyphosis (Scheuermann's kyphosis)
	Lumbar spine	Hyperlordosis (indicates sacroiliac joint disease)
	Foot arches	Pes planus (flat feet) and Pes cavus (high-arched feet)
	Toe clawing	Indicates plantar fascial fibromatosis
Behind	Shoulders	Tenderness
	Spine	Scoliosis (S-shaped spine)
	Iliac crest	ASIS symmetry Pelvic tilt
	Gluteal	Wasting of gluteal muscles
	Popliteal fossa	Baker's cyst (non-pulsatile) Popliteal aneurysm (pulsatile)
	Hind-foot	Thickening of the Achilles' tendon

6. Spine

- Look at the patient's spine for evidence of scoliosis, and from side for abnormal lordosis/kyphosis
- Assess **lateral flexion** of cervical spine: ask patient to tilt head to each side, moving their ear towards their shoulder
- Assess range of movement of the **TMJ** and deviation of jaw
- **Trigger squeeze supraspinatus** indicates whether the patient is any chronic pain
- **Schober's test**: assess lumbar flexion by placing a finger on two adjacent lumbar vertebral spines and ask the patient to touch their toes:
- Look for expansion on flexion and back together on extension (reduced flexion --> ankylosing spondylitis)

7. Arms

- h) Ask the patient to sit down on the couch and put their hands behind their head (tests **shoulder abduction**, **external rotation**, and **elbow flexion**)
- i) Ask patient to hold hands out, palms down and fingers outstretched to test for the extension of joints
- j) Assess backs of hand for **asymmetry**, **joint swelling** and **deformity**
- k) Gently squeeze across metacarpophalangeal joints and assess for signs of discomfort
- l) Ask patient to turn their hands over so their palms are facing up
- m) Assess muscle bulk of the palms for evidence of thenar/hypothenar wasting
- n) Ask patient to make a fist to test range of movement of small joints of fingers
- o) Assess **power** and **precision grip**

8. Legs

- p) Look at the quadriceps muscle bulk and assess for any swellings/deformities
- q) Perform patellar tap for knee effusion
- r) Assess **passive flexion** and **extension** of knee
- s) Assess **internal rotation** of hip
- t) Inspect feet and squeeze metatarsophalangeal joints to assess for pain- active inflammatory arthropathy

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

Bloods – Take a full blood count, U&Es, ESR, CRP, specific rheumatoid factors, and auto antibodies

Imaging – AP and lateral radiographs, MRI for soft tissue damage

Special tests – Joint aspiration and microscopy for crystals



Knee Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth?*
I have been asked to do an examination on your knee, which would involve me looking, feeling, and moving the knee joint. Is that ok?
- *Would you mind removing your trousers for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* mobility aids, Zimmer frame, shoes (heel raises and supports)
- *Look at the patient's hands discretely:* indicate OA or RA

3. Gait

- Ask the patient to walk over to the wall, turn around and walk back.
- Check for **symmetry**, **smoothness**, and **step height**
- Observe gait cycle (heel strike, toe-off). Is the turn smooth and quick?

Gait pathology	Sign	What it can indicate
Antalgic gait	Stance phase abnormally shortened to reduce time on that foot	Implies pain in specific leg
Ataxic gait	An unsteady, staggering gait because walking is uncoordinated	Cerebellar lesion
Festinant gait	Short accelerating steps often on tip-toes	Due to increased muscle tension in their legs commonly seen in Parkinson's disease
Fixed flexion/hyperextended knee	The knee is in fixed flexion/knee is hyperextended	Occurs in polio as patient have quadriceps wasting
High-stepping gait	Characterized by foot-drop due to loss of dorsiflexion	Damage to the deep fibular nerve
Trendelenburg's gait (look for Trendelenburg sign)	Support the patient's outstretched arms, get them to stand on one leg. If unsupported side of pelvis drops down, then positive sign.	Indicated damage to the gluteus medius on the supported side, due to lesion of the superior gluteal nerves

4. Look: Ask patient to stand and inspect the patient from the front, side and back

Action	Sign	What it may indicate
Front	Scars	Previous surgery
	Swelling	Presence of effusions around patellar: sub- and supra- patellar cysts Psoriatic plaques
	Asymmetry/leg length discrepancy	Growth arrest in childhood Patellar asymmetry
	Varus vs valgus	Fixed flexion/hyperextension Varus is more common and due to medial compartment osteoarthritis Valgus is more common in rheumatoid arthritis
	Quadriceps bulk	Wasting (sarcopenia)
Side	Foot deformity	Pes cavus (high arched foot) Pes planus (flat foot)

Behind	Asymmetry	Check if the iliac crests are level
	Hamstring bulk	Check if there is sarcopenia
	Popliteal swelling	Popliteal aneurysm Baker's cyst Semimembranosus cysts

5. **Feel:** Assess with patient reclined (~45 degree angle)

- Assess **temperature:** using back of hands, check temperature medially and laterally by patting down both legs. Compare knees against each other: increased temperature in inflammation
- Offer to measure **apparent** and **true** leg length
- If there is asymmetry, offer to measure **quadriceps circumference**
- Offer to assess:
 - a) **Lateral bulge test:** Swipe fluid from medial knee into suprapatellar pouch. Hold fluid there with one hand on medial side, and swipe down into lateral side with other hand. Medial sulcus obliteration is a positive test lateral --> suggest presence of effusion
 - b) **Patellar tap:** Milk fluid down from suprapatellar pouch (10cm above the patella). With other hand, place pressure on patella. A 'tap' heard on underlying femur --> indicates large joint effusion

Assess the joint line and feel:

- **Patella margins:** palpate medial and lateral patella facets for tenderness and swing patella side to side
- Medial and lateral aspects of the joint line
- **Quadriceps tendon insertion into patella:** tenderness indicative of tendonitis
- **Tibial tuberosity:** tenderness seen in Osgood-Schlatter disease
- **Head of the fibula:** identifying any head of fibula fractures
- With the knee lax at 30° feel for the **popliteal fossa**

6. **Move:**

ACTIVE MOVEMENT: Ask the patient to carry out a series of movements

- **Flexion:** ask patient to bend knee as much as possible (normal is 135°)
- **Extension:** ask patient to straighten their leg and extend their knee as much as possible
- Check if knee is **hyperextended:** ask patient to lift heel of bed (normal is 5°)

PASSIVE MOVEMENT: Ask patient to relax and allow you to move the joint freely taking the weight

- Passively assess **flexion and extension**
- **Hip internal rotation:** with knee bent internally rotate the leg --> Excludes hip disease
- **Hyperextension:** elevate both legs by the heels and note if there is any hyperextension (>10°)

7. **Special Tests:**

CRUCIATE LIGAMENTS: with knee flexed to 90 degrees

- **Anterior draw:** grasp upper tibia, with fingers in popliteal fossa and thumbs on tibial tuberosity, and pull forwards, abnormal anterior motion indicates loss of ACL.
- **Posterior Drawer test:** Push tibia posteriorly to assess PCL laxity

COLLATERAL LIGAMENTS: extend knee to 10 degrees

- For MCL—apply outwards pressure on foot, LCL—apply inwards pressure. Movement >5-10° is abnormal

MENISCUS: Offer McMurray's test to assess for medial and lateral meniscus tears

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

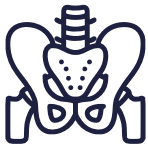
Bedside – (History) Take a full history

- **(Observations)** Examine neurovascular state of both limbs e.g. pulse, sensation, proprioception
- **(Corresponding examination)** Examine joint above and below (hip and ankle)

Imaging – AP and lateral radiographs of the knee

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Special tests – Offer McMurray's test to assess meniscal damage (can be painful to perform)



Hip Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an examination on your hip, which would involve me looking, feeling, and moving the hip joint. Is that ok?*

- *Would you mind removing your trousers for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* mobility aids, Zimmer frame, shoes (heel raises and supports)

3. Gait

- Ask the patient to walk over to the wall, turn around and walk back.
- Check for **symmetry**, **smoothness**, and **step height**
- Observe gait cycle (heel strike, toe-off)
- Is the turn smooth and quick?

Gait pathology	Sign	What it can indicate
Antalgic gait	Stance phase abnormally shortened to reduce time on that foot	Implies pain in specific leg
Waddling gait	Patient moves their upper body forwards and drags lower leg forward	Weakness of proximal muscles of pelvis which causes weakness of gluteus muscles
Spastic gait	Stiffness in the legs and tendency to circumduct the feet	Upper Motor Neuron (UMN) lesion
Fixed flexion/hyperextended knee	The knee is in fixed flexion/knee is hyperextended	Occurs in polio as patient have quadriceps wasting
High-stepping gait	Characterized by foot-drop due to loss of dorsiflexion	Damage to the deep fibular nerve
Trendelenburg's gait (look for Trendelenburg sign)	Support the patient's outstretched arms, get them to stand on one leg. If unsupported side of pelvis drops down, then positive sign.	Indicated damage to the gluteus medius on the supported side, due to lesion of the superior gluteal nerves

4. Patient standing

OBSERVE: Ask patient to stand and inspect the patient from the **front, side and back**

Action	Sign	What it may indicate
Front	Scars	Previous surgery
	Pelvic tilt	Weakness of muscles on one side
	Asymmetry/leg length discrepancy	Growth arrest in childhood
	Quadriceps wasting	Wasting in chronic joint disease
Side	Foot deformity	Pes cavus (high arched foot) Pes planus (flat foot)
	Loss of lumbar lordosis	Fixed flexion deformity
Behind	Gluteal wasting	
	Scoliosis	

	Asymmetry of pelvic brim- dimples of Venus	These are sagittal symmetrical indentations sometimes visible on the lower back, just superior to the gluteal cleft. These indentations are created by a short ligament stretching between the posterior superior iliac spine and the skin.
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- Perform **Trendelenburg Test**
 - a) Stand in front of the patient and ask them to put outstretched hands on your arms for support.
 - b) Look for weakness of hip abductors and depression on that side
- A positive sign (pelvic drops on the opposite side) suggests weakness of abductor muscles --> gluteus medius/minimus muscles

5. Patient reclined

OBSERVE: Ask patient to recline on the couch

- Assess **skin** for scars, pigmentation
- Assess **muscles** for wasting and fasciculation
- Offer to measure:

Apparent leg length on both sides	This is from the umbilicus to medial malleolus	Unequal suggests tilting pelvis, due to spinal/pelvic deformity
True leg length on both sides	This is from the ASIS to ipsilateral medial malleolus	Unequal suggests actual limb shortening e.g., fracture, hip disease

FEEL: With patient lying down, ask about pain and start with normal side first.

- Palpate **anterior aspect of hip**– Assess temperature, tenderness, and joint insertions --> Inflammation/infection
- Feel for **greater trochanter** – thumb on ASIS and move fingers down--> trochanteric bursitis

ACTIVE MOVEMENT:

- Place one hand under the lumbar spine to detect masking of hip movement by spine
- **Flexion:** bring knee towards chest (normal ROM is 120°)

PASSIVE MOVEMENT: Ask patient to relax and allow you to move the joint freely, roll each leg side to side to make the limb floppy

- **Flexion:** bring knee towards chest (normal ROM is 120°)
- **Internal and external rotation:** whilst the knee and hip are flexed to 90°:
 - a) Turn the knee **inwards** (Internal rotation --> normal ROM is 30°)
 - b) Turn the knee **outwards** (External rotation --> normal ROM is 40°)
- **Extension:** place one hand on pelvis and lift one leg at a time to assess extension (normal ROM is 10-20°)

6. Special tests

- **Thomas' test:** tests fixed flexion deformity (DO NOT perform on patients with hip replacement as can cause dislocation)
 - u) Place a hand under the patient's lumbar spine
 - v) Ask patient to bring both knees to chest to straighten the pelvis- lumbar lordosis should be flattened
 - w) Ask patient to hold one knee and extend the other- repeat on other leg
- A fixed flexion deformity will prevent straightening of the leg --> Suggestive of osteoarthritis
- A patient may compensate with increased lumbar lordosis i.e. spine lifting off from your hand

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Examine neurovascular state of both limbs e.g. pulse, sensation, proprioception
- (**Corresponding examination**) Examine joint above and below (knee and spine), check for the presence of hernias)

Imaging – AP and lateral radiographs

Special tests – Test resisted hip flexion, adduction, and sensation of antero-lateral thigh



Shoulder Examination

1. Introduction

- Wash Your hands
- Introduce yourself by name and role
- Check their identity – name and DOB
- Explain the procedure - why you need to do it and what does it involve
- Ask for consent
- Expose the patient appropriately
- Check if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. Can I check your name and date of birth? I have been asked to do an examination of your shoulder, which would involve me looking, feeling, and moving the joint. Is that ok?
- Would you mind removing your shirt for me please?
- And can I just check whether you are in any pain?

2. Bedside Inspection

- Observe the patient: Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- Observe the surroundings: "paraphernalia of MSK disease"

3. Patient standing

OBSERVE: Inspect the patient from the front, side and back, asking the patient to turn as you do so

Action	Sign	What it can indicate
Front	Scars	Previous surgery
	Asymmetry of shoulder girdle	Scoliosis Arthritis Trauma
	Swelling	Inflammatory joint disease
	Deltoid wasting	Axillary nerve injury
	Arm position	Internally rotation indicates posterior shoulder dislocation
Side	Scars	
Behind	Muscle bulk of trapezius	
	Back muscle bulk	Check if there is sarcopenia
	Scoliosis	
	Winged scapula	Damage to serratus anterior/long thoracic nerve

FEEL: Check pain first and start on the normal side

- Using back of hands, check **temperature** over both shoulders
- Assess the **shoulder girdle**
 - a) Feel sternoclavicular joint--> along the clavicle--> acromioclavicular joint
 - b) Palpate coracoid process
 - c) Feel head of humerus
 - d) Work around the glenohumeral joint
 - e) Start from spine of scapula working upwards back to the acromioclavicular joint
- Assess the muscle bulk of **supraspinatus**, **infraspinatus**, and **deltoid**
- Ask patient to flex the biceps and feel the tendon for biceps tendonitis

ACTIVE MOVEMENT:

- Quick screening test: arms above the head and behind the back
- Check neck movements: flexion, extension, turning head and tilting
- 'Can you raise your arm keeping it straight'
- **Flexion:** ask patient to raise arm forwards (normal is 180°)
- **Extension:** ask patient to swing arms back (normal is 65°)
- **Abduction:** raise each arm up sideways, ensure you hold the inferior pole of scapula
- **Adduction:** move arms across body (normal is 50°)
- **External rotation:** place patients arm flexed to 90°, then turn outwards
- **Internal rotation:** place hand on back and reach as far as possible (normal T4-T8)

PASSIVE MOVEMENT: If abnormalities noted on active movements

- Ask patient to relax and allow you to move the joint freely taking the weight, whilst feeling for crepitus
- Passively assess **flexion, extension, abduction, adduction, external** and **internal rotation**

4. Special tests

SUPRASPINATUS: Empty can test

- Flex shoulder to 90°, thumbs pointing down, bend elbow slightly, try and resist downward movement placed on their ulnar
- Tests for weakness or impingement of the supraspinatus tendon

INFRASPINATUS

- Resisted external rotation in neutral adduction
- Pain may suggest infraspinatus tendonitis

TERES MINOR

- Position the arm in 90° of abduction and bend the elbow to 90°
- Passively externally rotate the shoulder to its maximum degree

SUBSCAPULARIS

- Ask the patient to place the dorsum of their hand on their lower back
- Apply light resistance to the hand by pressing it towards their back
- Ask the patient to move their hand off their back
- Inability to do this indicates pathology of the subscapularis (e.g., tendonitis/tear)

SCARF TEST

- Put patient's hand over their contralateral shoulder
- Pain over acromioclavicular joint indicates osteoarthritis

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history, assess functional status, effect on the patient's activities of daily living, etc.

- (**Corresponding examination**) Examine the joint above and below (cervical spine and elbow), assess neurovascular state of both limbs (pulse, sensation, and proprioception)

Bloods – Would take a full blood count + U&Es and cardiac markers

Imaging – AP and lateral radiographs of the shoulder



Hand Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth?*
I have been asked to do an examination of your hand, which would involve me looking, feeling, and moving the joints in your hand. Is that ok?
- *Would you mind rolling up your sleeves for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* “paraphernalia of MSK disease”

3. Look

OBSERVE: Ask patient to stand and inspect the patient from the **front, side** and **back**

Action	Sign	What it may indicate
Dorsum of hand	Hand posture	
	Scars	Scoliosis Arthritis Trauma
	Erythema	Cellulitis Joint Sepsis Pallor
	Skin thinning	Long term steroid use
	Swelling	Axillary nerve injury
	Bouchard nodes	Osteoarthritis
	Heberden's nodes	
	Swan-neck deformity	Rheumatoid arthritis
	Boutonnières deformity	
	Z thumb	
	Ulnar deviation	
	Nail pitting	Psoriasis
	Onycholysis	
Palms up	Scars	
	Skin colour	
	Thenar/hypothenar wasting	Carpal tunnel syndrome
Elbow	Psoriatic plaques	Psoriasis

4. Feel

Hands: Check for pain and start on the normal side.

Feel the palms, facing upwards:

- Assess **temperature:** using back of hands, check temperature over wrists and small joints of hand
- Assess **radial and ulnar pulse**
- Assess **thenar/hypothenar eminence bulk**
- Feel for **palmar thickening**--> Dupuytren's contracture

Feel the palms, facing downwards:

- Assess **temperature:** using back of hands, check temperature over wrists and small joints of hand
- Squeeze metacarpophalangeal joints

- Bimanually palpate the joints of the hand: **MCP--> PIP--> DIP--> Carpometacarpal joint**
- Palpate **anatomical snuffbox** and feel wrists for tenderness

Elbows: Work up from ulnar border to the elbow

- Assess for any nodule or psoriatic plaques

SENSATION:

- Check **median nerve** sensation over thenar eminence then index finger
- Check **ulnar nerve** sensation over hypothenar eminence then little finger
- Check **radial** sensation over first dorsal web space

5. Move

ACTIVE MOVEMENT:

- **Flexion:** make a fist
- **Extension:** open fist and splay fingers
- **Wrist extension:** make your hands like you are praying
- **Wrist flexion:** put back of your hands together

PASSIVE MOVEMENT:

- Repeat movements passively

MOTOR ASSESSMENT: *Ask patient to carry out the following movements under resistance*

- **Finger extension-->** Radial nerve
- **Finger abduction-->** Ulnar nerve
- **Thumb abduction-->** Median nerve

FUNCTION:

- **Power grip:** *"squeeze my fingers with your hand"*
- **Pincer grip:** *"squeeze my finger between your thumb and index finger"*
- *"Pick up a coin or undo a shirt button"*

6. Special tests

- **Tinel's test: Used to assess for carpal tunnel syndrome**
 - x) Tap over the carpal tunnel with your finger
 - y) If the patient develops tingling in the thumb and radial 2 and ½ fingers, this is suggestive of median nerve irritation and compression
- **Phalen's test: Used to assess for carpal tunnel syndrome**
 - z) Ask the patient to hold their wrist in complete and forced flexion for 60s
 - aa) If the patient's symptoms of carpal tunnel syndrome are reproduced then the test is positive

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- **(Observations)** Examine neurovascular state of upper limbs e.g. pulse, sensation, proprioception
- **(Corresponding examination)** Examine joint above (elbow)

Imaging – AP and lateral radiographs of the hand



Cranial Nerves Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an examination of the nerves supplying the head, this will involve testing facial movements and sensation and checking hearing and vision*
- And can I just check whether you are in any pain?

2. Bedside Inspection

- **Observe the patient:** Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
 - o In addition check for rashes, wasting and scars (behind the ears)
- **Observe the surroundings:** mobility aids (wheelchair, walking stick), medications, glasses

3. Cranial Nerve I (Olfactory)

- Ask patient if they have noticed a change in smell recently. Can use a scented bottle (coffee/mint)

4. Cranial Nerve II (Optic)

- **Acuity:** Ask if patient wears glass (if yes, ask patient to wear them during assessment)
 - o Test one eye at a time: assess distant vision with a Snellen chart (Acuity = distance from chart/line read e.g. 6/60), assess near vision – ask patient to read a line of magazine. If unable, ask to count fingers → assess perception of movement (wave hand) → simple light perception
 - o Offer to test colour vision with Ishihara plates
- **Fields**

Test for	Action	Significance
Visual inattention	With both eyes open, waggle fingers of right hand then left then together. Ask patient to state which hand is moving.	Inattention to one side suggests a contralateral parietal lesion
Visual field mapping	Sit patient 1 meter opposite you + cover eye not tested - Move finger from periphery to centre in the 4 quadrants, staying in a horizontal plane	Monocular loss Bitemporal hemianopia Homonymous hemianopia (optic tract lesion)
Blind spot	Assess like fields but with a red pin, map the size of the blind spot	Large blind spot- papilloedema

- **Reflexes:**
 - o **Accommodation:** focus on distant then near object checking for pupil restriction + convergence
 - o **Pupil reflex:** Shine light in each pupil + observe for direct and consensual pupillary constriction
 - o **Swinging light test:** shine light between two eyes, pupil size should stay equal. If one eye dilates, relative afferent pupil defect- *Marcus-Gunn pupil* (partial optic nerve lesion)
- Offer **fundoscopy** to assess the optic disc

5. Cranial Nerve III (Oculomotor), IV (Trochlear), VI (Abducens)

- **Inspect:** **ptosis** (partial in Horner's syndrome, full in CN III lesion), **strabismus/squint** (manifest vs latent)
- **H-test:** patient's eyes follow your finger tracing a H shape. Pause laterally to check for **nystagmus** (cerebellar lesion.) If any double vision (**diplopia**) ask if vertical or horizontal and perform cover test
 - o CN XI lesion- loss of lateral rectus (eye adducted and cannot move laterally)
 - o CN IX lesion- loss of superior oblique (eye cannot move down when facing inwards, diplopia when looking down)
 - o CN III lesion- most eye movements impaired (eye rests down and out)

6. Cranial Nerve V (trigeminal)

- Inspect: temporalis and masseter wasting
- Sensory: use sternum as reference, ask if can feel and if feeling the same on both sides for supra-orbital notch, maxilla and chin. Touch both sides after to assess for sensory inattention (neglect.)
- Motor: ask patient to clench jaw and feel temporalis and masseter muscle bulks, open and close mouth against resistance
- Offer corneal reflex (afferent Va and efferent VII) and jaw jerk reflex

7. Cranial Nerve VII (Facial)

- Inspect: for facial symmetry (creases of forehead and face)
- Motor: raise eyebrows (spared in UMN lesion), screw up eyes, puff cheeks out, show teeth and grimace
- Ask if sounds are abnormally loud (hyperacusis = damage to stapedius)
- Offer to check taste sensation over anterior 2/3 of the tongue (chorda tympani)

8. Cranial Nerve VIII (Vestibulocochlear)

- Ask patient whether they use a hearing aid, and have noticed any abnormalities in hearing (tinnitus) or balance (vertigo)

Test	Significance
Crude hearing test- rustle fingers in alternative ear and whisper number into other ear	Indicates whether there is a hearing deficit in either ear
Weber's test- using a tuning fork, place on forehead and ask if one side is louder	If one side is louder then either that side has conductive deficit, or contralateral has sensorineural deficit
Rinne's test- using a tuning fork, place base on mastoid process and ask patient to tell you when sound stops	Check if louder in air or on bone. If bone louder indicates a conductive hearing loss.

- Offer Romberg's test of balance, vertigo tests like walking on the spot + Dix-Hallpike test.
- Offer inspection with auroscope if any previous test is abnormal

9. Cranial Nerve IX (Glossopharyngeal) and X (Vagus)

- **Inspect:** Assess palate symmetry (CN IX) and uvula deviation (CN X), deviates away from side of lesion
- **Motor:** Ask patient to cough and assess speech (impaired in recurrent laryngeal nerve palsy) and swallow (assesses muscles supplied by the pharyngeal plexus)
- Offer to test **gag reflex:** Sensory = CN IX, Motor = CN X

10. Cranial Nerve XI (Spinal Accessory)

- **Inspect:** SCM and trapezius for sign of wasting
- **Motor:** Ask patient to turn head against resistance (assess contralateral SCM muscle) and shrug shoulders against resistance (assess trapezius)

11. Cranial Nerve XII (Hypoglossal)

- **Inspect:** Assess tongue for wasting or fasciculations (LMN lesion e.g. due to bulbar palsy)
- **Motor:** Ask patient to move tongue from side to side (deviation to the side of lesion)

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- **(Observations)** Full set of observations
- **(Corresponding examination)** Conduct a full peripheral neurological assessment
- Assess speech and cognitive function. Assess functional status



Lower Limb Neuro Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain
- **Ask** if right or left-handed

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an examination of the nerves in your legs, this will involve having a look and feel of your legs and testing the movement and sensation in your legs. Is that ok?*

- *For the purposes of this examination I will need to see your whole legs, so would you mind undressing below the waist please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* mobility aids (wheelchair, walking stick), NG tube

3. Gait and screening tests

- Ask patient to stand up with arms folded – tests proximal weakness
- Walk around back and observe for any spinal scars, cutaneous signs
- Ask patient to walk – assess start (hesitant in PD), rate, gait, arm swinging and turn
- Ask patient to do heel-toe walking – if impaired signals, cerebellar lesion
- Stand on toes (tests S1, sciatic nerve power, plantarflexors) and heels (tests L4/5, impaired in footdrop)
- Assess balance using Romberg's test – Ask patient to put feet together and close eyes whilst standing
 - o Reduced stability with eyes closed (+ve sign) – indicates proprioceptive/dorsal column dysfunction.
 - o Reduced stability with eyes open (-ve sign) – indicates cerebellar ataxia

4. Inspection of the legs- lie the patient back down

Action	Signs	What it may indicate
Inspect patient's legs	S - Scars	Could indicate previous surgery, or trauma
	W - Wasting	Indicates loss of innervation to muscles
Look closely for plantar foot wasting and dorsal foot guttering (LMN lesion)	I - Involuntary movements	Indicates upper motor neuron lesions: Chorea – Huntington's disease, Myoclonus - central nervous system disorders, Athetosis - damage of thalamus, basal ganglia, Pseudoathetosis - disruption of the proprioceptive pathway, from nerve to parietal cortex.
Look for bony deformity (pes cavus)	F - Fasciculations	Indicates lower motor neuron lesions
	T -Tremor	Can be fine or course. Resting tremor found in Parkinson's
	P - Posture	Indicates curvature of the spine
	H - Hypertrophy	

5. Tone

- Ask patient to relax muscles so you can passively move them
- **Roll leg side to side:** Increased tone = UMN lesion, Decreased tone = LMN lesion
- **Lift knee up and down quickly:** Look for spasticity (foot kicks out involuntarily = UMN lesion)
- **Clonus- rotate and quickly dorsiflex:** Feel for rhythmic beats of gastrocnemius (>2 = UMN lesion)

6. Power

- Test one joint at a time and support the joint being tested.
- Assess using MRC grades: 5 = full power 4 = some resistance, 3 = Can support against gravity 2= gravity eliminated, 1 = flicker of muscle contraction, 0 = nothing

Action	What it tests
Hips- lift legs up with knee extended, stabilises contralateral hip	L _{2,3} – Hip flex – <i>iliopsoas</i> – lift up leg (femoral nerve) L _{4,5} – Hip extension – <i>gluteal muscles</i> – Press leg into bed (sciatic – inf gluteal) L _{2,3} – Hip abductors/adductors
Knee- knee flexed to 90 degrees	L _{3,4} – Knee extensors (femoral) – <i>quadriceps</i> – straighten knee (femoral) L ₅ , S ₁ – Knee flexors – <i>hamstrings</i> (sciatic)
Ankle- legs straight ankle plantar/dorsiflexed	L _{4,5} – Ankle extensors (dorsiflexion) – <i>tibialis anterior</i> (sciatic – common peroneal) S _{1,2} – Ankle plantar flexors – <i>gastrocnemius</i> (sciatic)
Foot- eversion and inversion	L ₅ – foot evertors – <i>peroneus longus and brevis</i> (peroneal) L ₅ – foot invertors – <i>tibialis posterior</i> (tibial)
Toes	L ₅ , S ₁ – Toe extensors – <i>extensor digitorum longus</i> S _{1,2} – Toe flexors – <i>flexor digitorum longus</i> L ₅ – Hallux extension – <i>extensor hallucis longus</i> (common peroneal) S _{2,3} – Hallux flexion – <i>flexor hallucis longus</i> (tibial)

7. Reflexes

- **Knee jerk** reflex (L_{3/4}, kick the door), **Ankle** (S_{1/2}, in the shoe), **Plantar Babinski (S₁)** – Scrape sole from down later up and across. If hyperreflexia = UMN lesion
- Can ask patient to close eyes and clasp hands and pull outwards to accentuate the reflex

8. Coordination

- **Heel shin test** – ask patient to touch knee to hand and then contralateral knee. Move heel down tibia and back up to hand. Repeat 3 times. Impairment = cerebellar lesion.

9. Sensation

- Ask the patient 2 questions throughout: Can you feel it on both sides, and do they feel the same?
- Assess **light touch** (with cotton wool or tapping) – assesses dorsal column neurons
- Assess responsiveness to **pain** stimuli using neurotip – assesses spinothalamic tract fibres
- Can offer to test responsiveness to **temperature** – ask for hot or cold object
- Touch sternum as a reference point, followed by subsequent dermatomes:

T12- ASIS	L4- medial leg
L1- high pocket	L5- dorsum of big toe
L2- lateral thigh	S1- sole of foot
L3- medial thigh/patella	S2- popliteal fossa
- Assess responsiveness to vibration (dorsal column) – twang tuning fork and place on sternum as a reference point. Hold it on interphalangeal joint of the hallux, if can feel stop there. If not, work more proximal, move to metatarsophalangeal joint → ankle → knee
- Assess proprioception: - hold proximal phalanx of hallux and explain moving up and down With eyes closed, move up/down >3 times and assess whether patient can tell direction. If fail, move to ankle and then knee.

Thank the patient and wash your hands again.

“To complete the examination, I would do a number of steps...”

Bedside – (History) Take a full history including effects on life

- **(Observations)** Full set of observations
- **(Corresponding examination)** Conduct a neurological assessment of upper limbs and cranial nerves. Offer a full assessment of cognitive and higher order functions like speech



Upper Limb Neuro Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain
- **Ask** if right or left-handed

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an examination of the nerves in your arm, this will involve having a look and feel of your hands and testing the movement and sensation in your arms. Is that ok?*

- *Would you mind removing your shirt for me please so I can see the arms, shoulders and neck fully?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* mobility aids (wheelchair, walking stick), NG tube

3. Inspection of the arms

Action	Signs	What it may indicate
Inspect resting position, raise arms to look in axilla, inspect from behind. Ask the patient to hold out their arms and close their eyes (allows to see tremor and involuntary movements)	S - Scars	Could indicate previous surgery, or trauma
	W - Wasting	Indicates loss of innervation to muscles
	I - Involuntary movements	Indicates upper motor neuron lesions: Chorea – Huntington's disease, Myoclonus - central nervous system disorders, Athetosis - damage of thalamus, basal ganglia, Pseudoathetosis - disruption of the proprioceptive pathway, from nerve to parietal cortex.
	F - Fasciculations	Indicates lower motor neuron lesions
	T - Tremor	Can be fine or coarse. Resting tremor found in Parkinson's
	P - Posture	Indicates curvature of the spine
	H - Hypertrophy	

4. Screening

- Ask patient to hold out arms fully extended with palms facing upwards + close eyes
- Assess **pronator drift** – due to pyramidal weakness, UMN lesion. Assess upward drift- cerebellar lesion.

5. Tone

- Ask patient to relax muscles so you can passively move them. One hand hold elbow and other use to give handshake
- Assess elbow flexion and extension, supinator catch and wrist flexion, extension, rotation

Sign	Significance
Supinator catch	Indicative of UMN lesions
Increased vs decreased tone	UMN vs LMN lesion
Cogwheel rigidity	Parkinson's disease (extrapyramidal disorders)

6. Power

- Apart from shoulders, test one side at a time, and support the joint being tested.
- Assess using MRC grades: 5 = full power 4 = some resistance, 3 = Can support against gravity 2= gravity eliminated, 1 = flicker of muscle contraction, 0 = nothing

Action	What it tests
Shoulders- "arms like a chicken"	C5 – Shoulder abduction – <i>deltoid/supraspinatus</i> (axillary) C6,7,8 – Shoulder adduction – <i>pectorals</i> (median and lateral pectoral nerve)
Elbows- "arms like a boxer"	C6 – Elbow flexion – <i>biceps</i> (musculocutaneous), <i>bracioradialis</i> (in mid-pronation, radial nerve) C7 – Elbow extension - <i>triceps</i> (radial)

Wrists- "push and pull against resistance"	C7,8 – Wrist flexion – <i>flexor extensor carpi groups</i> (C7 – median, C8 – ulnar) C7 - Wrist extension – <i>extensor carpi groups</i> (radial)
Fingers- "spread fingers"	C7 - Finger extension – <i>extensor digitorum groups</i> (post interosseous nerve) C8 – Finger flexion – <i>flexor digitorum profundus</i> (I+II – median; III+IV – ulnar) T1 - Finger abduction, try to push in little and index finger – <i>1st dorsal interossei</i> (DAB; ulnar n), Finger adduction, try to push out fingers with your abductors. <i>palmar interossei</i> (PAD; ulnar n) Can use Froment's test for ulnar palsy – grab paper between thumb and index – loss of adductor pollicis
Thumb- "thumbs pointing up"	T1 - thumb abduction- <i>abductor pollicis brevis</i> (median)

7. Reflexes

- **Biceps** (C5/6, musculocutaneous nerve), **Supinator** (C5/6, radial nerve), **Triceps** (C7/8, radial nerve)
- Can ask patient to close eyes and grit their teeth to accentuate the reflex.

8. Coordination

- **Finger nose test** – Look for intention tremor and past pointing (dysmetria) – shows cerebellar lesion
- **Dysdiadochokinesis** – Alternate palms and back of hand on the other palm quickly – impaired in cerebellar lesions.

9. Sensation

- Ask the patient 2 questions throughout: Can you feel it on both sides, and do they feel the same?
- Assess **light touch** (with cotton wool or tapping) – assesses dorsal column neurons
- Assess responsiveness to **pain** stimuli using neurotip – assesses spinothalamic tract fibres
- Can offer to test responsiveness to **temperature** – ask for hot or cold object
- Touch sternum as a reference point, followed by subsequent dermatomes:

C4- shoulder tip	C8- little finger (ulnar)
C5- axillary nerve region, lateral arm	T1- medial forearm
C6- thumb/first web space (musculocutaneous)	T2- medial arm
C7- middle finger	
- Assess responsiveness to vibration (dorsal column) – twang tuning fork and place on sternum as a reference point. Hold it on interphalangeal joint of thumb, if can feel stop there. If not, work more proximal, move to metacarpophalangeal joint → wrist → elbow
- Assess proprioception: - hold proximal phalanx of thumb and explain moving up and down With eyes closed, move up/down >3 times and assess whether patient can tell direction. If fail, move to wrist and then elbow.

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- **(Observations)** Full set of observations
- **(Corresponding examination)** Conduct a neurological assessment of lower limbs and cranial nerves and offer a functional assessment eg doing up a button, writing



Diabetic Foot Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a student doctor.
Can I just check your name and date of birth?

- I have been asked to do a diabetic foot examination; this would involve having a look at you walk, having a look at your feet and testing your sensation.

For this examination would you mind undressing from waist below keeping your underwear on.

- Before I start, can I check whether you are in any pain?

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* “any walking aids, special footwear, orthotics, blood sugar monitoring”

3. Gait – Ask the patient to walk over to the wall, turn around and come back.

Comment on swing, stance, heel strike, turn cycle and any other significant abnormalities

- Assess **balance** using Romberg’s test – Ask the patient to put feet together and close eyes whilst standing:
 - a) Reduced stability with eyes closed (+ve sign) – indicates proprioceptive/dorsal column dysfunction.
 - b) Reduced stability with eyes open (-ve sign) – indicates cerebellar ataxia

- Inspect the patient’s shoes --> look for soles and padding, worn tread.

4. Legs – Ask patient to sit on couch with legs outstretched

Inspection – look at front and back of legs comparing between them. Also inspect between the toes

What to Look for	What it can indicate
Scars	Vessel harvesting for previous cardiovascular surgery
Colour	Cyanosis/pallor signifies peripheral vascular disease
Loss of digits	May indicate previous critical ischaemia/gangrene
Ulcers	Describe margin, colour, wet/dry and location
Skin trophic changes	Associated with peripheral ischaemia
Charcot joint	Consequence of diabetic neuropathy

- Gross motor assessment – ask patient to wiggle toes --> indicates paralysis due to ischaemia

Palpation

- Temperature – Feel temperature with back of hand from toes upwards and then compare sides
- Capillary refill – should be <2s. Compare both sides --> delayed if poor peripheral circulation

Pulses – (ask to lower patient’s boxers) --> poor peripheral pulses suggest ischaemia/poor circulation

- Start with *femoral pulse*
- Assess the *popliteal pulse*
- *Posterior tibial* – found 1cm posterior to the medial malleolus of tibia --> confirm presence and compare sides
- *Dorsalis pedis* – found on dorsum of foot between 2nd/3rd/ cuneiforms --> confirm presence and compare sides

Sensation

- i) Assess with monofilament at pulp of hallux, then pulp of 3rd digit
 - Then assess metatarsophalangeal joints of 1,3 and 5
- ii) Assess vibration on distal interphalangeal joint of hallux
- iii) Assess proprioception of distal interphalangeal joint of hallux

Reflexes --> Ankle reflex

Thank the patient and wash your hands again.

On completion, I would like to do a full history, a full set of observations and peripheral vascular and lower limb neurological examination

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Full set of observations including blood pressure
- (**Corresponding examination**) Conduct a peripheral vascular and lower limb neurological examination
- (**Bedside tests**) Would do a point-of-care glucose measurement

⇒ HOW TO PRESENT YOUR DIABETIC FOOT EXAMINATION

▪ Summary

- Today I performed a diabetic foot examination on Mr/Miss... a [age] male/female
- Bedside examination revealed paraphernalia/no paraphernalia of endocrine pathology
- The patient appeared alert, with a normal body habitus, comfortable at rest
- A peripheral examination revealed (no) stigmata of diabetic disease/ There was evidence of...
- Inspection of the legs revealed...
- Palpation of the legs revealed...
- Examination of the pulses of the lower limb revealed
- The sensory examination was...
- The ankle reflex was...

▪ Diagnosis

- These findings are consistent with ...

▪ Investigations

- Re-affirm history, basic observations and peripheral vascular and LLN examination
- Bedside-tests
 - Urinalysis + U A:Cr
 - BM
 - Fundoscopy
 - ECG
- Bloods
 - FBC
 - U&E
 - A:Cr ratio
 - HbA1c
 - Lipid Profile
- Further tests (only if required)
 - Duplex USS and calculation of ABPI
 - CT or MR angiography



Peripheral Vascular Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their Identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do a peripheral vascular examination on you, which would involve me having a look at your hands, face and chest, and then having a feel and listen to your chest. Is that ok?*

- *Would you mind removing your shirt for me please?*
- *And can I just check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* wheelchair, walking aids, oxygen
- *Any obvious scars/cyanosis*

3. Arms

- **INSPECTION:** ask the patient to hold their arms out and inspect for several features
- **PALPATION:** feel for temperature and compare between both sides. Test capillary re-fill on both sides. Palpate
- **PULSES:** **Radial** (rate, rhythm, ask for radio-radio delay which may signify dissection or coarctation)
- **Brachial-** compare volume of both sides and ask for blood pressure bilaterally (should be <15mmHg difference)

Arm signs	What it can indicate
Scars	Vessel harvesting for previous surgery
Pallor/cyanosis	Peripheral vascular disease
Loss of digits	May indicate previous critical ischaemia/gangrene
Tar staining	Smoking
Skin changes eg hair loss, shiny skin, ulcers	Associated with peripheral ischaemia

4. Head and neck

- **INSPECTION:** assess for conjunctival pallor (anaemia) and assess eyes for corneal arcus and xanthelasma (hyperlipidaemia)
- **PULSES:** carotid, first auscultate for carotid bruits and then feel one by one (not together)

5. Chest and abdomen

- **INSPECTION:** lie the patient supine and assess chest for scars (midline sternotomy for CABG or valve replacement). Inspect the abdomen for a pulsatile mass (abdominal aortic aneurysm.)
- **PALPATION:** palpate for abdominal aortic pulse (expansile, pulsatile mass indicative on aneurysm)
- **AUSCULTATION:** above the umbilicus (aortic bruits) and 2cm lateral and superior to umbilicus (renal bruits)

6. Legs

- **INSPECTION:** inspect front and back of legs for scars, colour, loss of digits, tar staining and skin changes. Inspect also between the toes
- Perform a gross motor assessment: ask the patient to wiggle their toes (paralysis secondary to ischaemia)
- **PALPATION:** assess temperature and capillary refill
- **PULSES:** **Femoral** (presence and volume, radial-femoral delay and auscultate for bruits- femoral/iliac stenosis). Check for presence and compare between sides for- **popliteal** (assess with knee flexed to 45 degrees), **posterior tibial** (1cm posterior to the medial malleolus), **dorsalis pedis** (dorsum of foot between 2nd/3rd cuneiform bones)
- **SENSATION:** assess light touch starting distally and comparing between sides

7. Special tests

- **BUERGER'S TEST**

- Slowly elevate both legs simultaneously assessing for the angle at which they become white: normal is $>90^\circ$, ischaemia at $20-30^\circ$, severe ischaemia $<20^\circ$. Ideally hold for 2-3 mins, realistically 4s intervals for each 10°
- Once time has elapsed place legs at 90° over the edge of the bed. Normal legs turn pink immediately. Ischaemic leg- foot turns pink slowly but then becomes dark pink after ~2mins (reactive hyperaemia due to vasodilatation from hypoxia)

Thank the patient and wash your hands again

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- (**Observations**) Full set of observations including blood pressure
- (**Corresponding examination**) Offer to do a cardiovascular examination and lower limb neurological exam
- (**Bedside tests**) blood glucoses (BMs) and urine dip for reno-vascular disease/glucose in the urine

Special tests – Ankle-brachial pressure index (>0.9 is normal)

Types of ulcers

	Arterial	Venous	Diabetic
Why they occur	Atherosclerosis	Venous stasis	Glycosylation of ECM
Location	Lateral side of leg	Around malleoli	Sole of foot
Appearance	Regular margins Punched out appearance	Irregular margins Pale surrounding skin Pigmentation due to haemosiderin deposit	Penetrating ulcer on toes
Painful	Yes	Yes	No
Associations	Cold extremities Trophic changes: shiny skin and hair loss	Often peripheral oedema Warm skin	Reduced sensation in surrounding skin Infected nails (fungal infection)



Cerebellar Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth? I have been asked to do an examination of neurological function. This will involve testing eye movements, speech, coordination and assessing you walking. Can I first check whether you are in any pain?*

2. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
 - o *Assess for abnormal posture and scars*
- *Observe the surroundings:* walking aids, medications, hearing aids

3. Standing and Gait

- Ask patient to sit up with arms folded to test for **truncal ataxia** (evidence of midline lesion)
- Romberg's test: unsteady on eyes open
- Assess gait: ataxic, wide based and stagger towards the side of the lesion
- Heel-toe walking (tandem gait): very difficult and uncoordinated

4. Eye Movements

- Assess for **nystagmus** (jerky movements on looking towards the side of the lesion)

	DANISH signs mnemonic
D	Dysdiadochokinesis
A	Ataxia
N	Nystagmus
I	Intention tremor
S	Slurred speech
H	Hypotonia

5. Speech

- Ask patient to say 'British Constitution': assessing for **dysarthria** (jerky, loud, irregular separation of syllables)

6. Upper Limbs-

- Arm drift: drift on extending arms due to hypotonia of agonist muscles
- Assess tone in shoulders, arms and wrists
- Finger-nose test: **intention tremor** on approaching target and past-pointing (**dysmetria**)
- **Dysdiadochokinesis**: rapidly alternating movements, e.g. palm and back of hand

7. Lower Limbs

- Assess lower limb tone- hips, knees and ankles
- Knee-jerk reflex: assess with legs over the side of the bed. Looking for a pendular reflex where the motion is slower in rise and fall
- Heel-shin test of coordination, intention tremor and past pointing on finger-toe testing, dysdiadochokinesis on tapping foot, hypotonia

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- **(Observations)** Full set of observations
- **(Corresponding examination)** peripheral neurological exams and cranial nerves examination

Imaging- neuroimaging eg MRI head if suspicion of demyelinating disease or SOL

Special- hearing assessment (acoustic neuroma)

Causes of cerebellar dysfunction:

Pattern of signs	Causes
Unilateral Disease	Space occupying lesions eg tumours and abscesses ○ Cerebellopontine angle tumour: assess cranial nerves for impaired CN V, VII, VIII function and perform fundoscopy to look for papilloedema ○ Malignancy Ischaemia ○ Vascular disease: eg vertebrobasilar disease, can perform a peripheral vascular examination looking for carotid bruits Multiple sclerosis
Bilateral disease	Drugs eg phenytoin Alcohol Multiple sclerosis: perform peripheral neurological examinations Hypothyroidism: thyroid exam Friedrich's ataxia Trauma Large space occupying lesion or cardiovascular disease



Parkinson's Examination

1. Introduction

- **Wash** Your hands
- **Introduce** yourself by name and role
- **Check** their identity – name and DOB
- **Explain** the procedure - why you need to do it and what does it involve
- **Ask** for consent
- **Expose** the patient appropriately
- **Check** if the patient is currently in any pain

e.g. Good morning, my name is .. and I am a medical student. *Can I check your name and date of birth?*
Today I would like to ask you some questions about the symptoms that you have been experiencing and test the movement and sensation in your arms and legs. Is that okay?
Would you mind rolling up your sleeves?
And can I just check whether you are in any pain?

2. Focused questions

- What happened when you first presented with this condition. How is it affecting you?
- When is your tremor worst?
- Do you have any problems with: balance or co-ordination, doing up buttons and tying shoelaces, getting in and out of your car?

3. Bedside Inspection

- *Observe the patient:* Patient is **A – Alert** **B – (normal) Body habitus** **C – Comfortable at rest**
- *Observe the surroundings:* “paraphernalia of neurological disease” (mobility aids)
- Do they have a resting tremor?

4. Gait Ask the patient to walk over to the wall, turn around and come back

- Ask patient to stand up with arms folded – tests proximal weakness
- Ask patient to walk – swing, stance, heel strike, turn cycle and any other significant abnormalities
- In the gait check for symmetry, smoothness, step height and turn (is it smooth and quick)
- **Shuffling gait** (reduced stride length), **Hesitant** (difficulty initiating and turning), **Festinating** (patient walks faster and faster to not fall over), **Lack of arm swing** (occurs occasionally in PD due to increased tone), **Unsteadiness** (tendency to fall forward and backwards), **Stooped posture**

5. Inspection of the face

- **Face:** Inspect for hypomimia (“mask face” - blank expressionless face with less blinking)
- **Eyes:** Glabella tap (tap on the forehead for Myerson’s sign, abnormal and a sign of frontal release)
- **Speech:** Ask patient to describe the room (hypophonia, slow thinking, soft, faint, hard to understand)

6. Tremors

- Resting tremor: typical ‘pill-rolling’ appearance, asymmetrical, 4-6 Hz in amplitude
- Postural Tremor: occurs during the maintenance of a position against gravity and worsens during active movement.
- Kinetic Tremor: perform the finger-nose test- simple kinetic tremor (remains constant through movement), intention tremor (worsens as the patient approaches the target)

7. Focused upper limb

- **Tone:** Ask patient to relax muscles so you can passively move them. Assess elbow flexion and extension, supinator catch and wrist flexion, extension, rotation
- An activation maneuver can accentuate subtle rigidity associated with early Parkinson’s – ask the patient to actively tap their thigh with their contralateral arm whilst you perform the movement.

Sign	Significance
Supinator catch	Indicative of UMN lesions
Increased vs decreased tone	UMN vs LMN lesion
Cogwheel rigidity	Parkinson’s disease (extrapyramidal disorders)

- **Bradykinesia:**
- Finger tapping (ask the patient to oppose their thumb and forefinger repeatedly, as fast as they can)
- Hand grip (ask the patient to make a fist and open their hand wide repeatedly, as fast as they can)
- Pronation/supination (ask the patient to pronate and supinate their hand repeatedly, as fast as they can.)
- Whilst observing the patient perform one or several of the above movements observe for the following: progressive reduction in **speed**, progressive reduction in **amplitude**, **asymmetry**, **slowness** in the initiation of movement

8. Focused lower limb

- **Bradykinesia:**
- Toe tap - Ask the patient to keep heel on the ground and tap their toes against the floor whilst seated as fast as they can

9. Extras

- Writing - Ask the patient to write a sentence and draw a spiral to assess for asymmetric progressive micrographia
- Buttons - Ask the patient to undo and do up their top shirt button (if present) to assess dexterity and speed of movement.
- Parkinson-plus syndromes: assess eye movements by doing the H Tracking for eye movements
 - o If problems in up and down --> progressive supranuclear palsy
 - o If problems in side to side with nystagmus --> multisystem atrophy

Thank the patient and wash your hands again.

"To complete the examination, I would do a number of steps..."

Bedside – (History) Take a full history

- **(Observations)** Full set of observations, in lying and standing blood pressure

Cerebellum: Perform a cerebellar examination

Eye movements: Assess eye movements for progressive supranuclear palsy (PSP).

Cognition: Perform a cognitive assessment (e.g. MMSE)

Drug chart: Analyze the drug chart for secondary parkinsonism